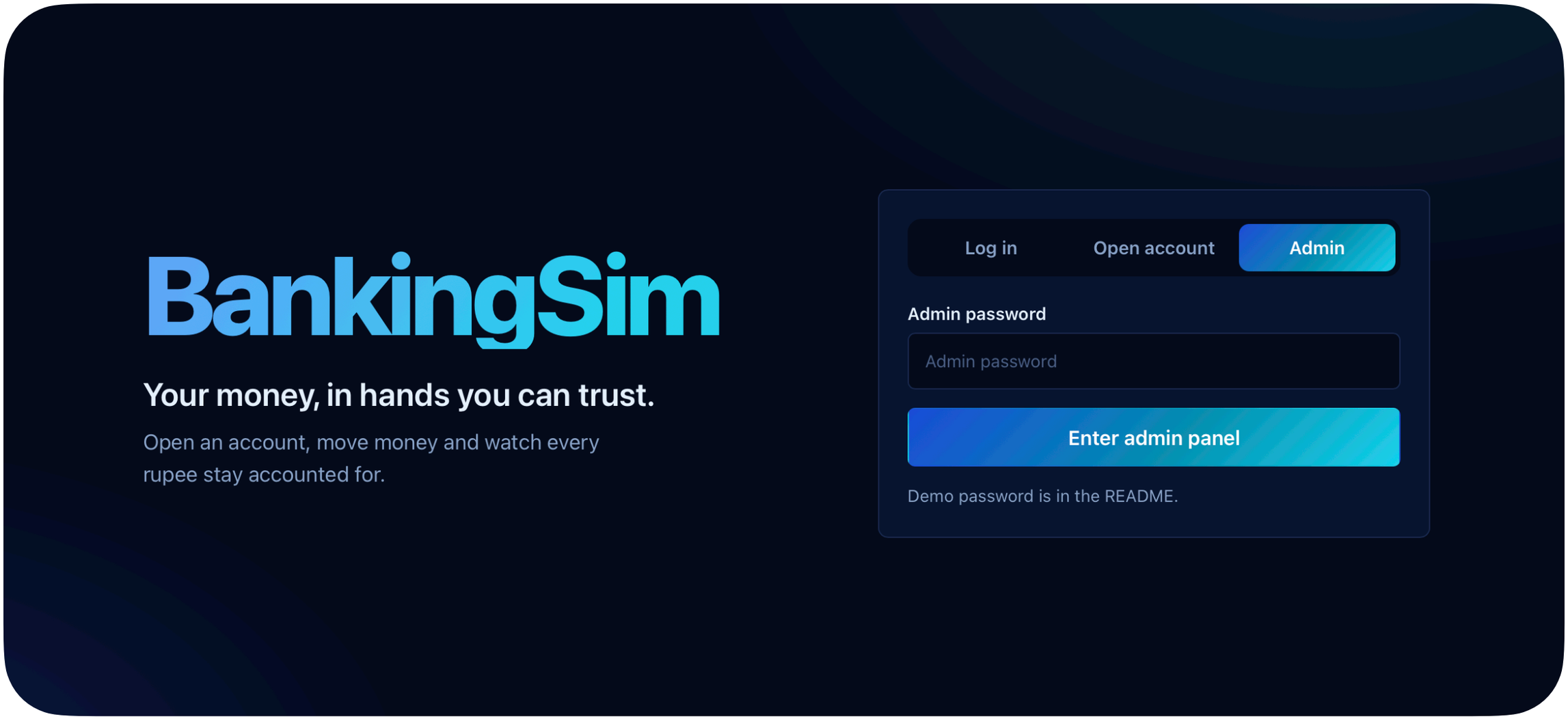


BankingSim - Java Assignment - 2

By Mohd Afnan, 1601257330 45



About BankingSim :

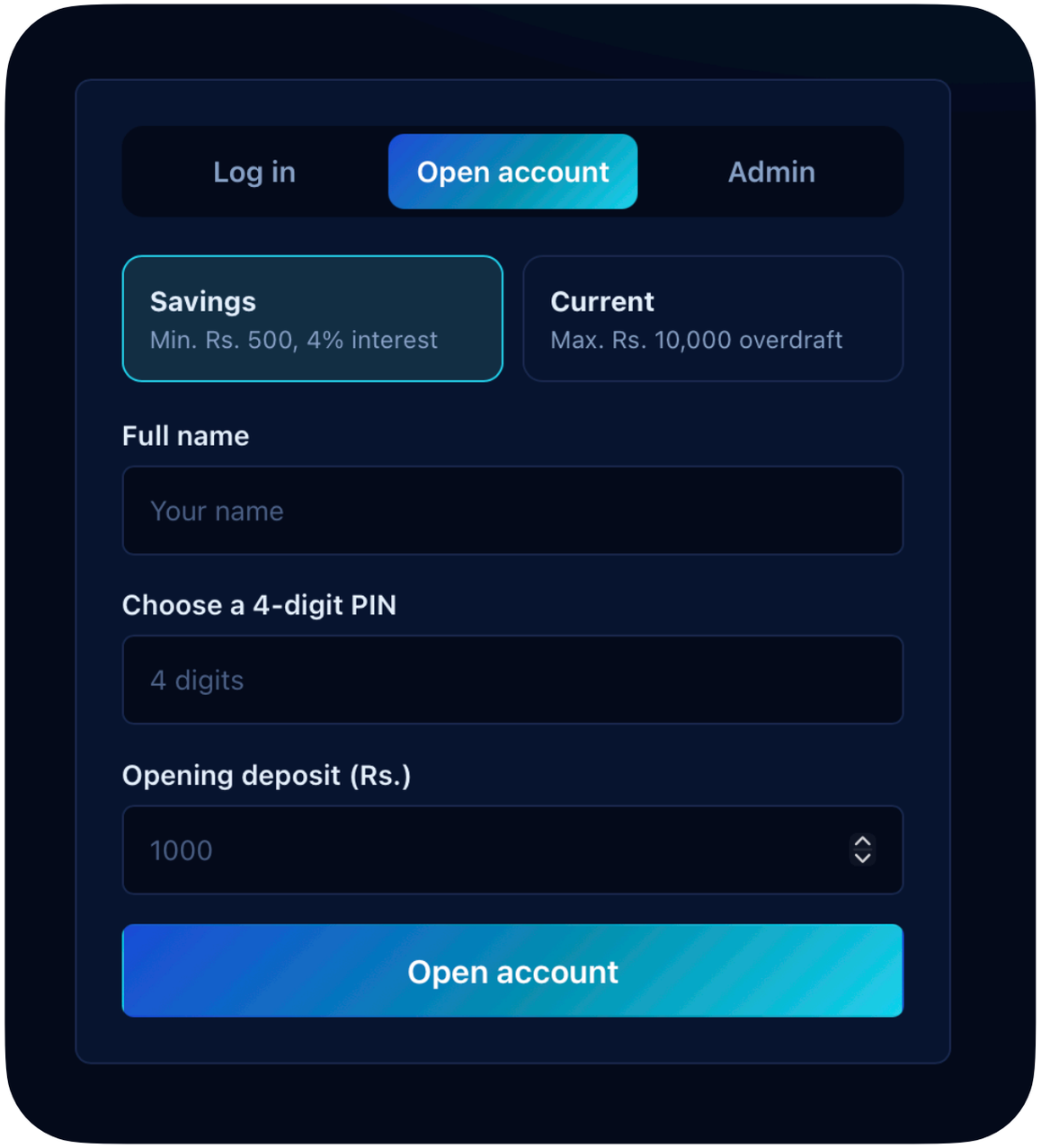
BankingSim is a Java banking simulator where customers open accounts, move money and review their history, and an administrator manages the bank. It started as a console program and now runs in the browser, with balances that stay correct under simultaneous use.

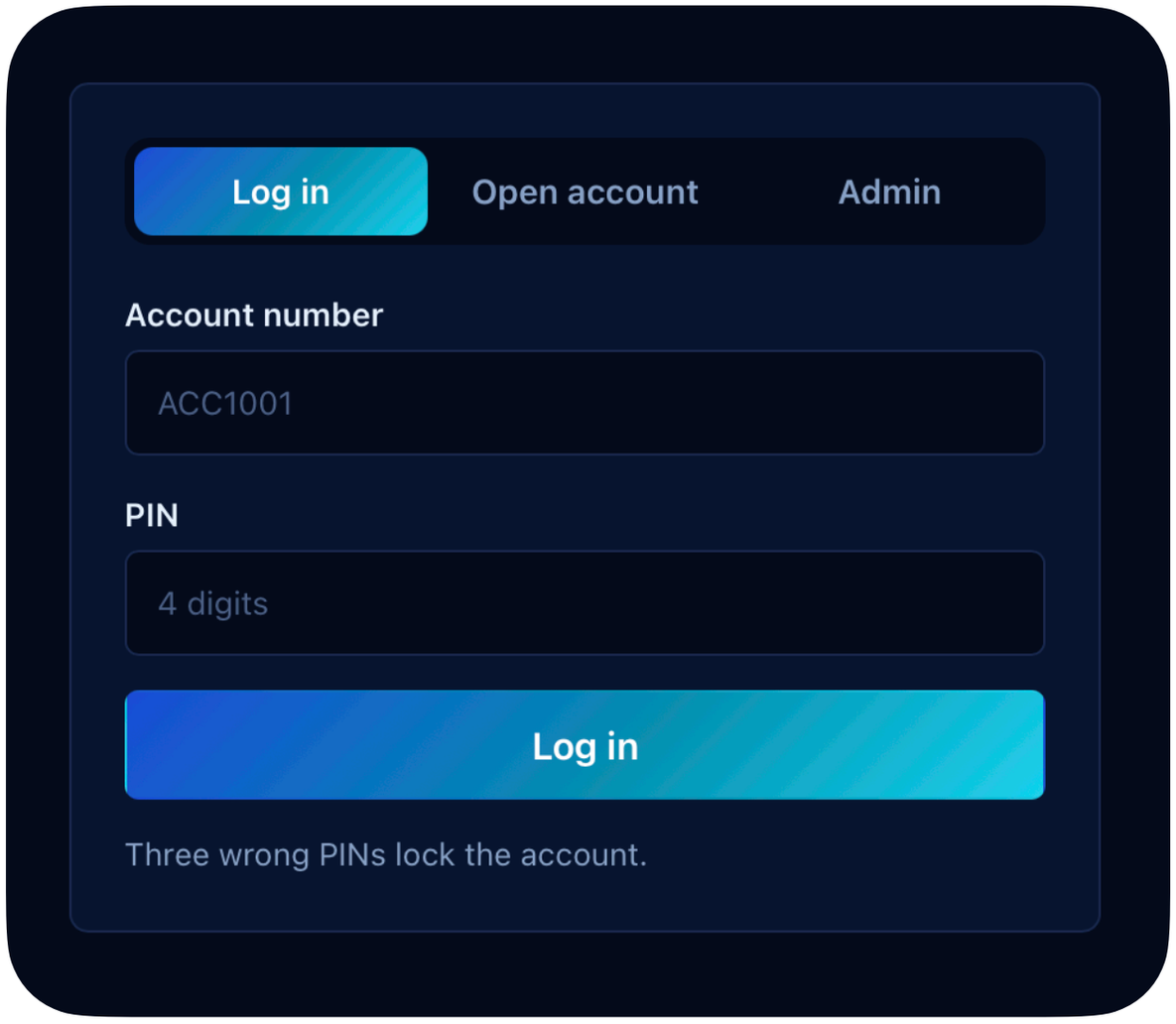
Key Details :

- **Account types:** Savings (minimum balance Rs. 500, 4% yearly interest paid monthly) and Current (overdraft up to Rs. 10,000).
- **Customer features:** open an account, log in with a 4-digit PIN, deposit, withdraw, transfer to another account, change PIN, and view or download transaction history.
- **Security:** three wrong PINs lock the account. Customers can only reach their own account, and admin actions need a separate admin login.
- **Admin features:** view totals and account mix, filter accounts by balance, lock or unlock accounts, pay interest, start an interest scheduler, create backups, and run a thread-safety demo.
- **Thread safety:** balance-changing methods are synchronised. Transfers lock accounts in a fixed order, so two opposite transfers can never deadlock.
- **Error handling:** custom exceptions for insufficient funds, invalid amounts, wrong PINs, locked accounts and unknown accounts.

Opening an Account :

A customer opens an account by choosing Savings or Current, entering their name, setting a 4-digit PIN and making an opening deposit (Rs. 500 minimum for Savings, Rs. 1,000 for Current). The bank then creates the account with a unique number such as ACC1001, records the deposit as the first transaction, and saves it. The customer is taken to the login tab with the new account number already filled in.



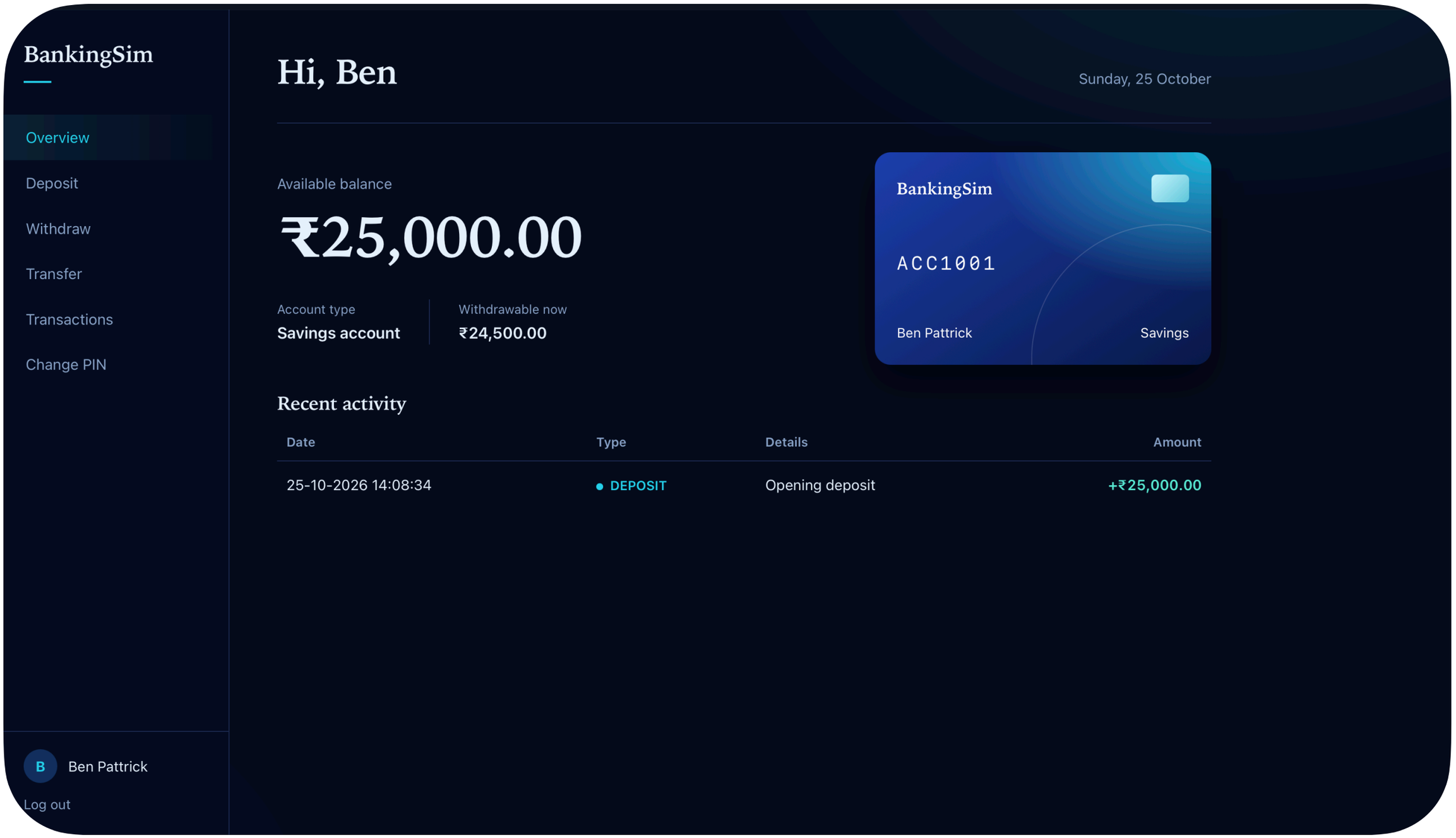


Logging In :

A customer logs in with their account number and 4-digit PIN. If the PIN is correct, the bank starts a session that lasts until they log out or stay idle for 30 minutes, and they land on their dashboard, which greets them by name. Each wrong PIN counts as a failed attempt and shows how many attempts remain. The third wrong PIN in a row locks the account until an admin unlocks it, and a correct PIN resets the count. Once logged in, a customer can only see and use their own account.

Difference between Accounts :

	Savings	Current
Opening deposit	Rs. 500 Min.	Rs. 1000 Min.
Balance rule	Must stay at or above Rs. 500	Can go into overdraft, down to -Rs. 10,000
Interest	4% a year, paid monthly	None
Best for	Saving money	Everyday spending



A Savings customer sees the same clean dashboard as everyone else, with a left-hand menu and a greeting by name, but the account type shows as “Savings” on the bank card and in the overview. The “Withdrawable now” figure is the balance minus the Rs. 500.minimum, and the monthly interest appears in the activity list as an Interest entry.



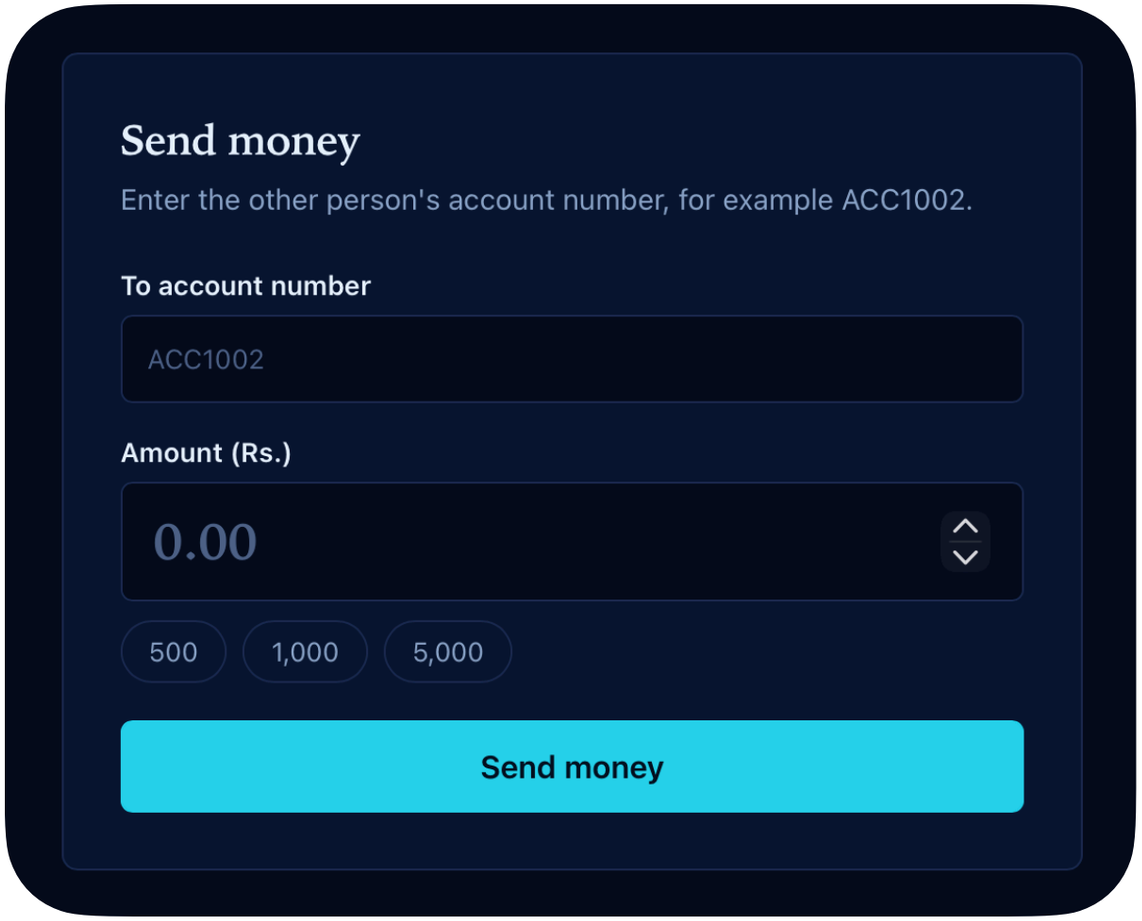
Deposit :

The customer enters an amount greater than zero, or taps a quick-fill button (500, 1,000 or 5,000), and the money is added to their balance straight away. The new balance shows on the dashboard and the deposit is recorded in the transaction history.

Withdrawal :

The customer enters an amount, and the bank allows it only if the account's balance rule is met. A Savings account must keep at least Rs. 500, and a Current account can go into overdraft down to -Rs. 10,000. If the amount is too high, the customer sees a message telling them the most they can withdraw. Locked accounts can't deposit or withdraw.

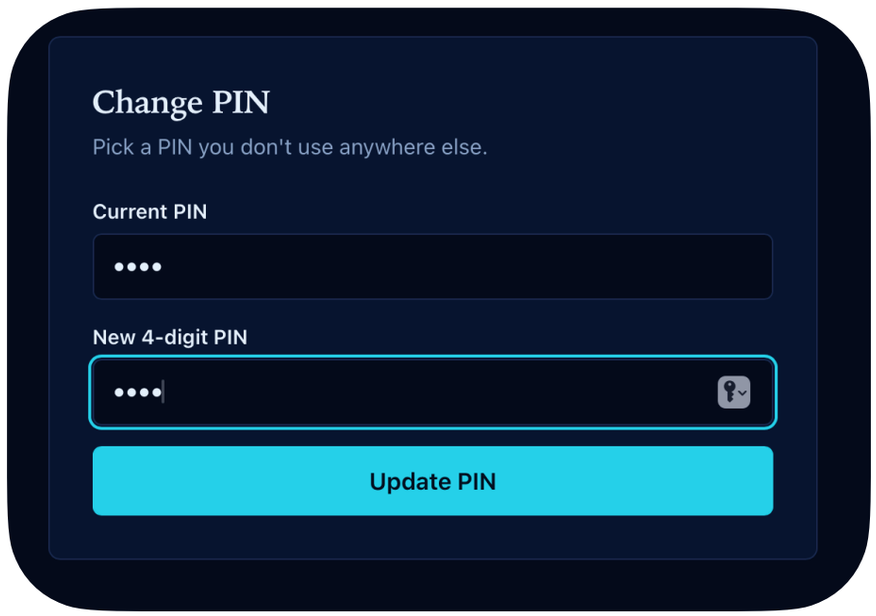


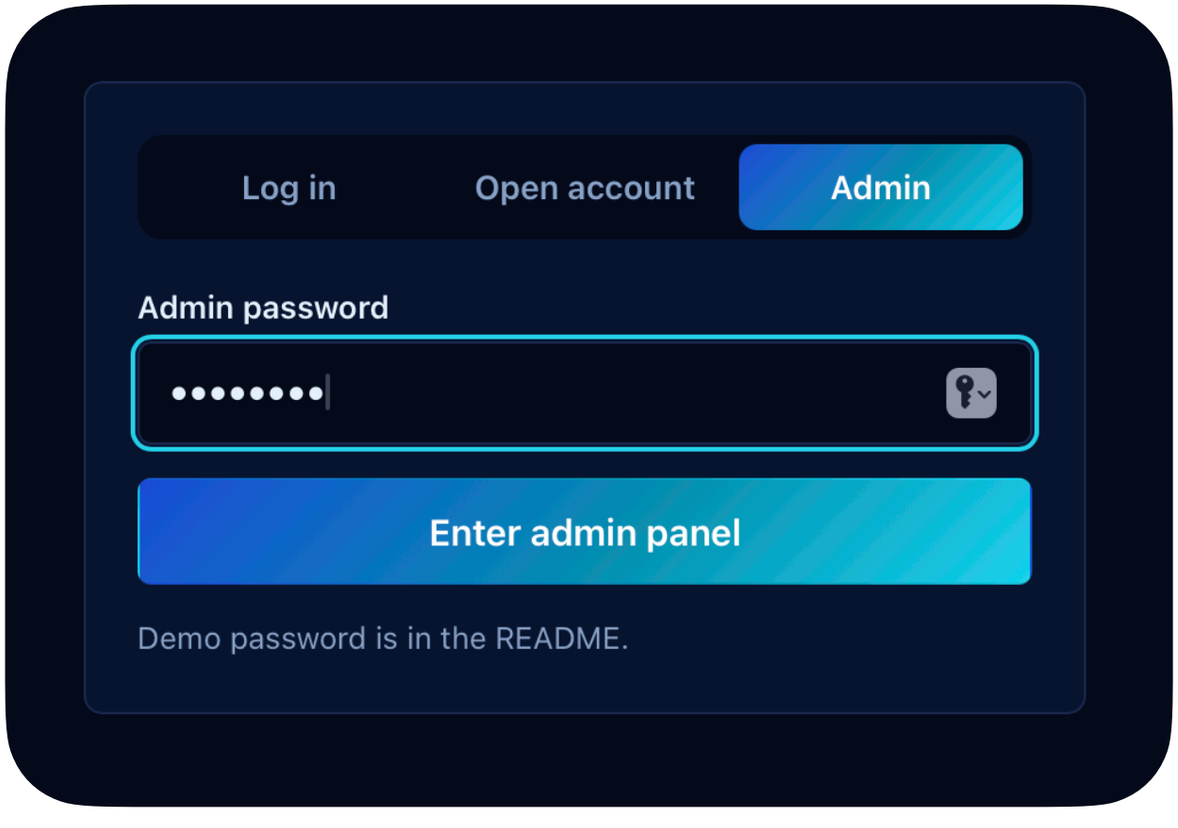


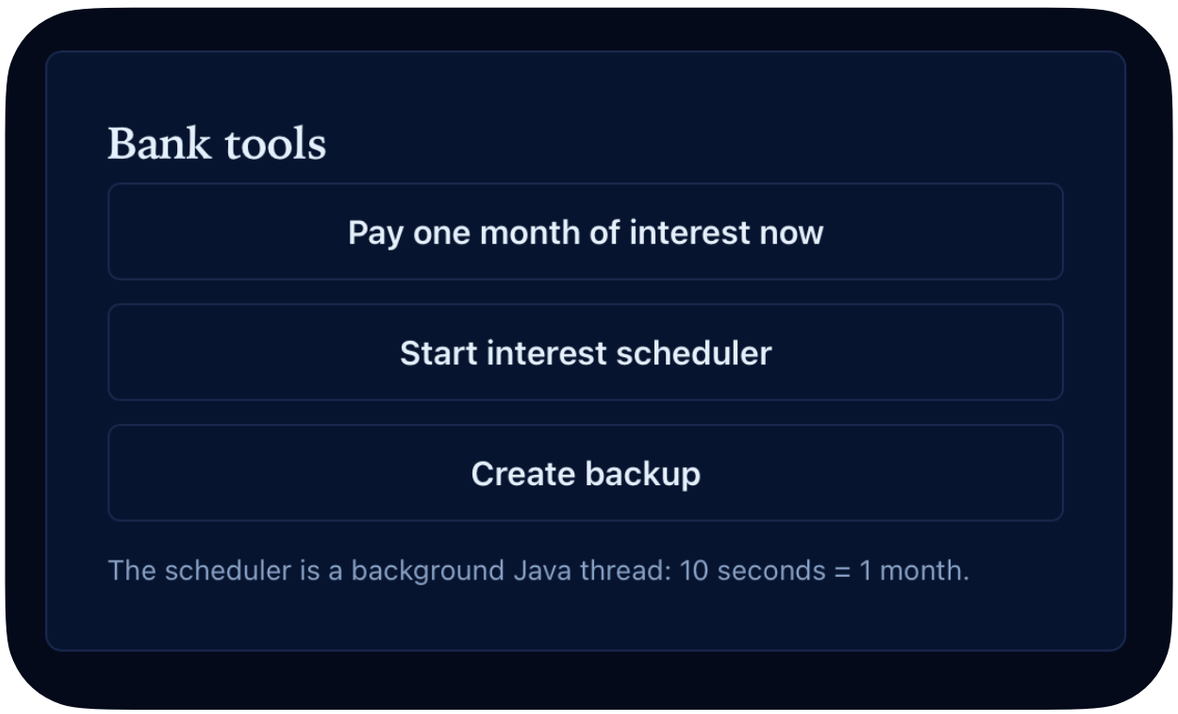
Other features :

A customer can **transfer** money to another account by entering its number and an amount; the transfer is refused if the amount is too high, the account doesn't exist, or either account is locked. They can **change their PIN** by entering the current PIN and a new 4-digit one. The **transaction history** lists every operation with its date, amount and balance afterwards, can be sorted, and can be downloaded as a CSV statement.









Administrator :

The administrator signs in from the Admin tab with a single password and gets a separate panel for managing the whole bank. The overview shows total accounts, total holdings, the Savings and Current split, and the highest balance, and the accounts list can be filtered by balance. The admin can lock or unlock any account, pay one month of interest to all unlocked Savings accounts now or through a background scheduler, and create a timestamped backup of all data. A built-in thread-safety demo runs real Java threads on throw-away accounts, so no customer data is touched.

Conclusion :

BankingSim shows that a small, careful Java core can power a real-feeling banking app. Savings and Current accounts follow their own rules, balances stay correct even when many operations happen at once, and custom exceptions turn mistakes into clear messages. The same core that ran in the console now serves a clean web interface for customers and an administrator. All accounts and transaction history are saved to a data file when the server stops and loaded again at the next start, with timestamped backups available from the admin panel, so nothing is lost between runs.